

The Full Asymptotic Transseries of the Swing Factorial and Its Inverse

Gamma-Ratio Normal Forms, Bernoulli–Bell Coefficient Calculus,
Borel Geometry, and Lambert-Normalized Reversion

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Abstract

The swinging factorial (or swing factorial)

$$\text{sf}(n) = \frac{n!}{\lfloor n/2 \rfloor!^2}$$

is not governed by a single smooth asymptotic scale: its even and odd subsequences have different algebraic prefactors, and the complete sequence falls sharply after every sufficiently large odd term. We therefore treat parity as a genuine two-component transseries sector. With $\sigma = -1$ for the even phase and $\sigma = +1$ for the odd phase, the canonical analytic branches are

$$\mathcal{S}_\sigma(x) = \frac{\Gamma(x+1)}{\Gamma(x/2 + (3-\sigma)/4)^2} = \frac{2^x}{\sqrt{\pi}} \left(\frac{\Gamma(x/2 + 1)}{\Gamma(x/2 + 1/2)} \right)^\sigma.$$

They admit the exact normal form

$$\mathcal{S}_\sigma(x) = C_\sigma 2^x x^{\sigma/2} \exp(\sigma D(x)), \quad C_\sigma = \frac{2^{-\sigma/2}}{\sqrt{\pi}},$$

where

$$D(x) = \frac{1}{2} \int_0^\infty e^{-xt} \frac{\tanh(t/2)}{t} dt.$$

Thus the Borel transform is the elementary meromorphic function $\widehat{D}(t) = \tanh(t/2)/(2t)$. This gives, simultaneously, the all-orders logarithmic expansion, an exact signed remainder, its optimal truncation scale, and the complete Borel singularity lattice.

The logarithmic coefficients are

$$d_{2r+1} = \frac{(2^{2r+2} - 1)B_{2r+2}}{(2r+1)(2r+2)}, \quad d_{2r} = 0.$$

Exponentiation gives every coefficient of the swing factorial itself by a complete exponential Bell polynomial, an explicit partition sum, or a triangular recurrence. For inversion, the dominant equation $L = \lambda x + (\sigma/2) \log x$, $\lambda = \log 2$, is solved exactly by Lambert's W -function. The real large branches require W_0 in the odd sector but W_{-1} in the even sector. Around that Lambert core the inverse has the constant-coefficient expansion

$$\mathcal{I}_\sigma(y) \sim X_\sigma(y) + \sum_{n \geq 1} \frac{c_n^{(\sigma)}}{X_\sigma(y)^n},$$

and we give both a Lagrange–Bürmann coefficient formula and an ordinary Bell-polynomial recurrence for arbitrary $c_n^{(\sigma)}$. A second, fully expanded power–logarithmic form

$$\mathcal{I}_\sigma(y) \sim \frac{1}{\lambda} \left(L_\sigma - \frac{\sigma}{2} H_\sigma + \sum_{n \geq 1} \frac{P_n^{(\sigma)}(H_\sigma)}{L_\sigma^n} \right)$$

is developed in parallel. General formulae for the polynomials $P_n^{(\sigma)}$ are expressed through coefficient extraction and ordinary Bell polynomials; their pure Lambert block is written explicitly with unsigned Stirling numbers of the first kind. We finish with Borel–Stokes transport under inversion, rigorous residual error estimates, numerical validation, and symbolic-generation code.

Keywords. Swinging factorial; central binomial coefficient; gamma ratio; Bernoulli numbers; Bell polynomials; Stirling numbers; Borel summation; resurgent asymptotics; Lambert W ; Lagrange–Bürmann inversion; power–logarithmic transseries.

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1 Introduction and statement of the main results

The sequence OEIS A056040 begins

$$1, 1, 2, 6, 6, 30, 20, 140, 70, 630, 252, 2772, \dots$$

and is commonly called the *swinging factorial*. Its elementary formulae, generating functions, and arithmetic interpretations are recorded in OEIS and in Luschny’s work on swinging factorials [1, 3]. The

existing OEIS entry A219931 also records a finite portion of the asymptotic expansion of its logarithm [2]. The purpose of this article is to put the problem into a complete coefficient-calculus and transseries framework, and then to carry out the substantially subtler inverse problem.

Two features dictate the organization.

1. The even and odd subsequences are respectively

$$\text{sf}(2m) = \binom{2m}{m}, \quad \text{sf}(2m+1) = (2m+1) \binom{2m}{m}.$$

Their leading powers of the index are opposite: $n^{-1/2}$ versus $n^{+1/2}$.

2. The full sequence is not eventually monotone. Indeed

$$\frac{\text{sf}(2m+1)}{\text{sf}(2m)} = 2m+1, \quad \frac{\text{sf}(2m+2)}{\text{sf}(2m+1)} = \frac{2}{m+1}.$$

Consequently there is no canonical one-valued inverse of the unsplit integer sequence.

The correct asymptotic object is therefore a pair of monotone analytic branches. Throughout,

$$\sigma = -1 \quad \text{means even phase}, \quad \sigma = +1 \quad \text{means odd phase}, \quad \lambda := \log 2.$$

At an integer n , its phase is

$$\sigma_n = (-1)^{n+1} = -(-1)^n.$$

It is useful to regard calculations as taking place in the parity algebra

$$\mathcal{R} = \mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1).$$

This records both sectors in one formula without pretending that parity is a small perturbation.

Forward transseries

For every fixed $N \geq 1$, uniformly in closed subsectors of $\Re x > 0$,

$$\mathcal{S}_\sigma(x) = C_\sigma 2^x x^{\sigma/2} \left(\sum_{n=0}^{N-1} \frac{\sigma^n A_n}{x^n} + \mathcal{O}(x^{-N}) \right), \quad C_\sigma = \frac{2^{-\sigma/2}}{\sqrt{\pi}},$$

where

$$A_n = \frac{1}{n!} Y_n(1!d_1, 2!d_2, \dots, n!d_n) = \sum_{\substack{k_1, k_2, \dots \geq 0 \\ \sum_{j \geq 1} j k_j = n}} \prod_{j \geq 1} \frac{d_j^{k_j}}{k_j!},$$

$$d_{2r+1} = \frac{(2^{2r+2} - 1)B_{2r+2}}{(2r+1)(2r+2)}, \quad d_{2r} = 0.$$

Equivalently,

$$A_0 = 1, \quad nA_n = \sum_{j=1}^n j d_j A_{n-j}.$$

The logarithmic series is Borel summable to an exact elementary Laplace integral, and its remainder after N nonzero terms has the sign of the first omitted term and smaller magnitude.

Lambert-normalized inverse transseries

Put

$$L_\sigma(y) = \log \frac{y}{C_\sigma} = \log y + \frac{1}{2} \log \pi + \frac{\sigma}{2} \log 2.$$

The exact inverse of the dominant block $L_\sigma = \lambda x + (\sigma/2) \log x$ is

$$X_{\sigma,k}(y) = \frac{\sigma}{2\lambda} W_k \left(2\sigma \lambda e^{2\sigma L_\sigma(y)} \right).$$

For the large positive real inverse,

$$X_+(y) = \frac{W_0(4\pi\lambda y^2)}{2\lambda}, \quad X_-(y) = -\frac{W_{-1}(-4\lambda/(\pi y^2))}{2\lambda}.$$

Then

$$\mathcal{I}_\sigma(y) = \mathcal{S}_\sigma^{-1}(y) \sim X_\sigma(y) + \sum_{n \geq 1} c_n^{(\sigma)} X_\sigma(y)^{-n}.$$

Define

$$Q_\sigma(t) = \sigma \sum_{j \geq 1} d_j t^j, \quad b = \frac{\sigma}{2}, \quad \Psi_\sigma(t, u) = -\frac{1}{\lambda} \left[b \log(1 + tu) + Q_\sigma \left(\frac{t}{1 + tu} \right) \right].$$

The general coefficient is

$$c_n^{(\sigma)} = \sum_{m=1}^n \frac{1}{m} [t^n u^{m-1}] \Psi_\sigma(t, u)^m,$$

where the notation means ordinary coefficient extraction in both variables. A completely triangular Bell-polynomial recurrence is given in Theorem 7.1.

Direct power-logarithmic inverse

Let

$$H_\sigma(y) = \log \frac{L_\sigma(y)}{\lambda}.$$

Then

$$\mathcal{I}_\sigma(y) \sim \frac{1}{\lambda} \left[L_\sigma - \frac{\sigma}{2} H_\sigma + \sum_{n \geq 1} \frac{P_n^{(\sigma)}(H_\sigma)}{L_\sigma^n} \right],$$

where $P_n^{(\sigma)}$ is a polynomial of degree at most n , generated at arbitrary order by Theorem 8.1. If the gamma-ratio correction is suppressed, the remaining pure Lambert polynomial has the closed form

$$P_n^{(0)}(H) = \frac{b^{n+1}}{n!} \sum_{m=1}^n (-1)^{m+1} (m-1)! \begin{bmatrix} n \\ m \end{bmatrix} \binom{n}{m-1} H^{n-m+1}, \quad b = \frac{\sigma}{2}.$$

Thus the inverse coefficients sit exactly at the intersection of the Bernoulli, Bell, and Stirling coefficient calculi.

2 Elementary structure and the two analytic branches

2.1 Equivalent definitions

Definition 2.1 (Swing factorial). For $n \in \mathbb{N}_0$, define

$$\text{sf}(n) := \frac{n!}{\lfloor n/2 \rfloor!^2}. \quad (2.1)$$

Writing $n = 2m$ or $2m + 1$ immediately gives

$$\text{sf}(2m) = \binom{2m}{m}, \quad \text{sf}(2m + 1) = (2m + 1) \binom{2m}{m}. \quad (2.2)$$

It also follows that

$$\text{sf}(0) = 1, \quad \text{sf}(n) = \begin{cases} n \text{ sf}(n-1), & n \text{ odd}, \\ \frac{4}{n} \text{ sf}(n-1), & n \text{ even}. \end{cases} \quad (2.3)$$

The product form in the OEIS definition is recovered by iterating Equation (2.3).

2.2 Generating functions

Proposition 2.2 (Ordinary and exponential generating functions). *The swing factorial has exponential generating function*

$$\sum_{n \geq 0} \text{sf}(n) \frac{z^n}{n!} = (1 + z)I_0(2z), \quad (2.4)$$

and ordinary generating function

$$\sum_{n \geq 0} \text{sf}(n) z^n = \frac{1}{\sqrt{1-4z^2}} + \frac{z}{(1-4z^2)^{3/2}} = \frac{1 + z/(1-4z^2)}{\sqrt{1-4z^2}}. \quad (2.5)$$

Proof. From Equation (2.2),

$$\sum_{m \geq 0} \text{sf}(2m) \frac{z^{2m}}{(2m)!} = \sum_{m \geq 0} \frac{z^{2m}}{(m!)^2} = I_0(2z),$$

and the odd exponential series is $zI_0(2z)$. For the ordinary series, use

$$F(u) := \sum_{m \geq 0} \binom{2m}{m} u^m = (1-4u)^{-1/2}.$$

The odd part is

$$z \sum_{m \geq 0} (2m+1) \binom{2m}{m} z^{2m} = z(2uF'(u) + F(u))|_{u=z^2} = \frac{z}{(1-4z^2)^{3/2}}.$$

□

The two singularities $z = \pm 1/2$ in Equation (2.5) have the same modulus. That co-dominance is the generating-function origin of the parity sector: no single nonoscillatory singular expansion can represent the full sequence. Even/odd projection separates the two contributions exactly.

2.3 Canonical gamma continuations

Definition 2.3 (Parity-resolved analytic branches). For $\sigma \in \{-1, +1\}$, define

$$\mathcal{S}_\sigma(x) := \frac{\Gamma(x+1)}{\Gamma(x/2 + (3-\sigma)/4)^2}. \quad (2.6)$$

Thus $\mathcal{S}_-(2m) = \text{sf}(2m)$ and $\mathcal{S}_+(2m+1) = \text{sf}(2m+1)$.

The duplication formula gives the more symmetric identity

$$\mathcal{S}_\sigma(x) = \frac{2^x}{\sqrt{\pi}} \left(\frac{\Gamma(x/2 + 1)}{\Gamma(x/2 + 1/2)} \right)^\sigma. \quad (2.7)$$

It follows directly from Equation (2.6) that

$$\frac{\mathcal{S}_\sigma(x+2)}{\mathcal{S}_\sigma(x)} = \frac{16(x+1)(x+2)}{(2x+3-\sigma)^2} = \begin{cases} \frac{4(x+1)}{x+2}, & \sigma = -1, \\ \frac{4(x+2)}{x+1}, & \sigma = +1. \end{cases} \quad (2.8)$$

The recurrence makes the common exponential scale 4 per two steps transparent.

Proposition 2.4 (Monotonicity and real inverses). *Both branches are strictly increasing for $x > 0$. More precisely,*

$$\frac{d}{dx} \log \mathcal{S}_\sigma(x) = \lambda + \frac{\sigma}{2} [\psi(x/2 + 1) - \psi(x/2 + 1/2)]. \quad (2.9)$$

The odd branch is strictly increasing on $[0, \infty)$; the even branch has zero derivative at 0 and positive derivative thereafter. Consequently each has a unique large positive inverse, denoted $\mathcal{I}_\sigma$.

Proof. The digamma difference in brackets is positive because ψ is increasing. Its derivative is $\frac{1}{2}[\psi_1(x/2 + 1) - \psi_1(x/2 + 1/2)] < 0$, since the trigamma function is positive and decreasing. At $x = 0$ the difference equals $2 \log 2$. Thus the derivative in Equation (2.9) is positive for $\sigma = +1$, and for $\sigma = -1$ it is zero only at $x = 0$ and positive for $x > 0$. $\square$

A global interpolation such as

$$\tilde{\mathcal{S}}(x) = \frac{1 + \cos \pi x}{2} \mathcal{S}_-(x) + \frac{1 - \cos \pi x}{2} \mathcal{S}_+(x)$$

matches every integer value, but is highly oscillatory and has no canonical one-valued large inverse. The discontinuous interpolation obtained by retaining $\lfloor x/2 \rfloor$ is no better. The phase-resolved pair $(\mathcal{S}_-, \mathcal{S}_+)$ is therefore not merely convenient; it is the asymptotically stable notion of continuation and inversion.

3 Exact logarithmic normal form

3.1 Separating the dominant scale

Set

$$D(x) := \log \frac{\Gamma(x/2 + 1)}{\Gamma(x/2 + 1/2)} - \frac{1}{2} \log \frac{x}{2}. \quad (3.1)$$

Then Equation (2.7) becomes the exact factorization

$$\mathcal{S}_\sigma(x) = C_\sigma e^{\lambda x} x^{\sigma/2} e^{\sigma D(x)}, \quad C_\sigma = \frac{2^{-\sigma/2}}{\sqrt{\pi}}. \quad (3.2)$$

At integer arguments this gives the exact parity form

$$\log \text{sf}(n) = n\lambda - \frac{1}{2} \log \pi + \sigma_n \left(\frac{1}{2} \log \frac{n}{2} + D(n) \right), \quad \sigma_n = (-1)^{n+1}. \quad (3.3)$$

The periodic multiplier is part of the leading transseries data, not an error term.

3.2 An exact Laplace representation

Theorem 3.1 (Elementary Borel–Laplace representation). *For $\Re x > 0$,*

$$D(x) = \frac{1}{2} \int_0^\infty e^{-xt} \frac{\tanh(t/2)}{t} dt = \int_0^\infty e^{-xt} \hat{D}(t) dt, \quad \hat{D}(t) = \frac{\tanh(t/2)}{2t}. \quad (3.4)$$

The apparent singularity of $\hat{D}$ at $t = 0$ is removable, with $\hat{D}(0) = 1/4$.

Proof. Let

$$h(z) = \log \Gamma(z+1) - \log \Gamma(z+1/2) - \frac{1}{2} \log z, \quad z = x/2.$$

Using the standard integral representation for a digamma difference and the identity $1/(2z) = \int_0^\infty e^{-2zu} du$, one obtains

$$\begin{aligned} h'(z) &= \psi(z+1) - \psi(z+1/2) - \frac{1}{2z} \\ &= - \int_0^\infty e^{-2zu} \tanh(u/2) du. \end{aligned}$$

Both $h(z)$ and the proposed integral tend to zero as $\Re z \rightarrow +\infty$. Integrating the derivative from infinity to z , then replacing $2z$ by x , gives Equation (3.4). Alternatively, differentiation under the integral sign verifies the same differential equation and the same boundary condition. $\square$

Several useful inequalities are immediate. Since $0 < \tanh(t/2) < t/2$ for $t > 0$,

$$0 < D(x) < \frac{1}{4x} \quad (x > 0). \quad (3.5)$$

Moreover $D'(x) < 0$, and complete monotonicity follows:

$$(-1)^m D^{(m)}(x) > 0 \quad (m = 0, 1, 2, \dots; x > 0). \quad (3.6)$$

3.3 The complete Borel singularity lattice

The partial-fraction expansion of the hyperbolic tangent gives

$$\hat{D}(t) = 2 \sum_{k=0}^{\infty} \frac{1}{t^2 + \pi^2(2k+1)^2}. \quad (3.7)$$

Indeed both sides are even meromorphic functions, have the same poles and residues, and agree at the origin. Thus the Borel singularities are the simple poles

$$t = \pm i\pi(2k + 1), \quad k = 0, 1, 2, \dots, \quad (3.8)$$

with residue $1/t$ at each pole. The nearest singularities are at $\pm i\pi$, so the formal expansion is Gevrey-1 with singulant modulus π .

Remark 3.2 (What “full transseries” means on the positive axis). The positive real Laplace ray meets no Borel singularity, so Equation (3.4) is unambiguous and equals the exact function. The quantity $e^{-\pi x}$ that appears in optimal truncation is the envelope set by the distance π of the nearest off-axis Borel poles. It should not be confused with an additional ambiguous positive-axis sector. Actual Stokes jumps arise only when the Laplace direction is rotated through the imaginary Borel rays; this is treated in Section 10.

4 The logarithmic asymptotic series and exact remainder

4.1 General coefficient formula

We use Bernoulli numbers with convention

$$\frac{z}{e^z - 1} = \sum_{n \geq 0} B_n \frac{z^n}{n!}, \quad B_1 = -\frac{1}{2}. \quad (4.1)$$

Theorem 4.1 (All logarithmic coefficients). *As $x \rightarrow \infty$ in any closed subsector of $|\arg x| < \pi/2$,*

$$D(x) \sim \sum_{r=0}^{\infty} \frac{d_{2r+1}}{x^{2r+1}}, \quad d_{2r} = 0, \quad (4.2)$$

where

$$d_{2r+1} = \frac{(2^{2r+2} - 1)B_{2r+2}}{(2r+1)(2r+2)} \quad (4.3)$$

$$= (-1)^r \frac{2(2r)!}{\pi^{2r+2}} (1 - 2^{-2r-2}) \zeta(2r+2). \quad (4.4)$$

Equivalently, the formal Borel transform is the Taylor expansion

$$\hat{D}(t) = \sum_{r \geq 0} d_{2r+1} \frac{t^{2r}}{(2r)!} = \frac{\tanh(t/2)}{2t}. \quad (4.5)$$

Proof. Expand each summand in Equation (3.7) at the origin:

$$\frac{1}{t^2 + a^2} = \sum_{r=0}^{N-1} \frac{(-1)^r t^{2r}}{a^{2r+2}} + \frac{(-1)^N t^{2N}}{a^{2N}(t^2 + a^2)}.$$

Termwise Laplace integration of the finite sum gives

$$d_{2r+1} = 2(-1)^r (2r)! \sum_{k=0}^{\infty} \frac{1}{\pi^{2r+2} (2k+1)^{2r+2}},$$

which is Equation (4.4). Euler’s formula for $\zeta(2r+2)$ yields Equation (4.3). The same calculation proves the asymptotic statement, while the unexpanded last term gives the exact remainder below. $\square$

The beginning of the series is

$$D(x) \sim \frac{1}{4x} - \frac{1}{24x^3} + \frac{1}{20x^5} - \frac{17}{112x^7} + \frac{31}{36x^9} - \frac{691}{88x^{11}} + \frac{5461}{52x^{13}} - \dots \quad (4.6)$$

Substitution in Equation (3.3) recovers and extends the logarithmic expansion recorded in OEIS A219931.

4.2 A signed next-term remainder theorem

Theorem 4.2 (Exact remainder and strict next-term bound). *For $x > 0$ and $N \geq 0$, write*

$$D(x) = \sum_{r=0}^{N-1} \frac{d_{2r+1}}{x^{2r+1}} + R_N(x). \quad (4.7)$$

Then, with $a_k = \pi(2k + 1)$,

$$R_N(x) = 2(-1)^N \sum_{k=0}^{\infty} a_k^{-2N} \int_0^{\infty} \frac{e^{-xt} t^{2N}}{t^2 + a_k^2} dt, \quad (4.8)$$

and

$$0 < (-1)^N R_N(x) < \frac{|d_{2N+1}|}{x^{2N+1}}. \quad (4.9)$$

Proof. Insert the finite geometric decomposition used in the proof of Theorem 4.1 into Equation (3.7), then into the exact Laplace integral. Every integral in Equation (4.8) is positive. Replacing $(t^2 + a_k^2)^{-1}$ by the strict upper bound a_k^{-2} , and evaluating $\int_0^{\infty} e^{-xt} t^{2N} dt = (2N)! x^{-2N-1}$, gives exactly the magnitude of the next term. $\square$

This theorem is stronger than a bare Poincaré estimate: every finite truncation brackets $D(x)$ with its successor. It also provides a simple certificate for all numerical calculations in later sections.

4.3 Large order and optimal truncation

From Equation (4.4),

$$d_{2r+1} \sim (-1)^r \frac{2(2r)!}{\pi^{2r+2}}. \quad (4.10)$$

The ratio of successive nonzero term magnitudes is asymptotic to $(2r + 2)(2r + 1)/(\pi^2 x^2)$. Hence the least term occurs near

$$r \sim \frac{\pi x}{2}. \quad (4.11)$$

Stirling's formula then gives the least-term envelope

$$\frac{2\sqrt{2}}{\pi\sqrt{x}} e^{-\pi x} (1 + \mathcal{O}(x^{-1})). \quad (4.12)$$

By Theorem 4.2, truncation at or immediately before the least term attains an error of this scale without any hidden constant explosion.

5 Forward transseries and Bell-polynomial coefficients

5.1 Exponential coefficient calculus

Let Y_n denote the complete exponential Bell polynomial, characterized by

$$\exp \left(\sum_{j \geq 1} x_j \frac{t^j}{j!} \right) = \sum_{n \geq 0} Y_n(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (5.1)$$

Theorem 5.1 (All forward coefficients). *Define $A_0 = 1$ and, for $n \geq 1$,*

$$A_n = [t^n] \exp \left(\sum_{j \geq 1} d_j t^j \right) \quad (5.2)$$

$$= \frac{1}{n!} Y_n(1!d_1, 2!d_2, \dots, n!d_n) \quad (5.3)$$

$$= \sum_{\substack{k_1, k_2, \dots \geq 0 \\ \sum_{j \geq 1} j k_j = n}} \prod_{j \geq 1} \frac{d_j^{k_j}}{k_j!}. \quad (5.4)$$

Then

$$\mathcal{S}_\sigma(x) \sim C_\sigma 2^x x^{\sigma/2} \sum_{n \geq 0} \frac{\sigma^n A_n}{x^n}. \quad (5.5)$$

The coefficients obey the triangular recurrence

$$A_0 = 1, \quad nA_n = \sum_{j=1}^n j d_j A_{n-j} = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} j d_j A_{n-j}. \quad (5.6)$$

Proof. Exponentiating Equation (4.2) and using Equation (5.1) gives Equation (5.3); expanding the exponential into a product gives the partition sum. Differentiating $A(t) = \exp(\sum d_j t^j)$ yields $A'(t) = (\sum j d_j t^{j-1}) A(t)$, and coefficient comparison proves Equation (5.6). Finally, because $D(t)$ is an odd formal series,

$$\exp(-D(t)) = \exp(D(-t)) = \sum_{n \geq 0} (-1)^n A_n t^n.$$

This explains the factor σ^n in Equation (5.5). □

Table 1: Universal coefficients A_n in $\exp D$.

n	A_n	n	A_n
0	1	7	$-39325/262144$
1	$1/4$	8	$-334477/8388608$
2	$1/32$	9	$28717403/33554432$
3	$-5/128$	10	$59697183/268435456$
4	$-21/2048$	11	$-8400372435/1073741824$
5	$399/8192$	12	$-34429291905/17179869184$
6	$869/65536$	13	$7199255611995/68719476736$

5.2 The two explicit forward sectors

The even branch is

$$\begin{aligned} \mathcal{S}_-(x) \sim \sqrt{\frac{2}{\pi}} 2^x x^{-1/2} & \left(1 - \frac{1}{4x} + \frac{1}{32x^2} + \frac{5}{128x^3} - \frac{21}{2048x^4} - \frac{399}{8192x^5} \right. \\ & \left. + \frac{869}{65536x^6} + \frac{39325}{262144x^7} - \frac{334477}{8388608x^8} + \dots \right). \end{aligned} \quad (5.7)$$

At $x = 2m$ this is exactly the classical central-binomial expansion in the variable $2m$.

The odd branch is

$$\mathcal{S}_+(x) \sim \frac{2^x x^{1/2}}{\sqrt{2\pi}} \left(1 + \frac{1}{4x} + \frac{1}{32x^2} - \frac{5}{128x^3} - \frac{21}{2048x^4} + \frac{399}{8192x^5} + \frac{869}{65536x^6} - \frac{39325}{262144x^7} - \frac{334477}{8388608x^8} + \cdots \right). \quad (5.8)$$

For the integer sequence, combine these as

$$\text{sf}(n) \sim \frac{2^n}{\sqrt{\pi}} 2^{-\sigma_n/2} n^{\sigma_n/2} \sum_{k \geq 0} \frac{\sigma_n^k A_k}{n^k}, \quad \sigma_n = (-1)^{n+1}. \quad (5.9)$$

This is a periodic or two-phase transseries: the discrete transmonomial σ_n has magnitude one and must remain visible at every order.

5.3 Gevrey character and Borel summability

The exact Borel representation proves that the logarithmic series is 1-summable in every direction that avoids the rays through $\pm i\pi(2k+1)$. Exponentiation preserves Gevrey-1 character and sectorial summability. Therefore Equation (5.5) is not merely a formal manipulation of Stirling series: in the right half-plane it is the asymptotic expansion of the exact gamma-ratio branch Equation (2.6). For fixed N ,

$$\mathcal{S}_\sigma(x) = C_\sigma 2^x x^{\sigma/2} \left(\sum_{n=0}^{N-1} \frac{\sigma^n A_n}{x^n} + \mathcal{O}_N(x^{-N}) \right). \quad (5.10)$$

After near-least-term truncation, the relative error has the same $e^{-\pi x}/\sqrt{x}$ envelope as the logarithmic correction, although its hyperasymptotic prefactor is modified by exponentiation.

6 A universal coefficient calculus for exponential–power inversion

The inverse calculation is clearer if the swing factorial is placed inside a slightly more general class. Let $a > 0$, b be fixed, and suppose

$$F(x) = C e^{ax} x^b \exp Q(x^{-1}), \quad Q(t) = \sum_{j \geq 1} q_j t^j. \quad (6.1)$$

At the formal level Q may be any series over a commutative $\mathbb{Q}$ -algebra. Analytically it may be a Gevrey asymptotic series with a chosen sectorial sum. Taking logarithms gives

$$L := \log(y/C) = ax + b \log x + Q(x^{-1}). \quad (6.2)$$

The swing branches correspond to

$$a = \lambda, \quad b = \frac{\sigma}{2}, \quad q_j = \sigma d_j. \quad (6.3)$$

6.1 A parameterized Lagrange lemma

Lemma 6.1 (Lagrange fixed-point formula). *Let R be a commutative $\mathbb{Q}$ -algebra and $\Psi(t, u) \in tR[[t, u]]$. The equation*

$$U(t) = \Psi(t, U(t)) \quad (6.4)$$

has a unique solution $U \in tR[[t]]$, and

$$[t^n]U(t) = \sum_{m=1}^n \frac{1}{m} [t^n u^{m-1}] \Psi(t, u)^m. \quad (6.5)$$

Proof. Introduce an auxiliary marker z and solve $U = z\Psi(t, U)$ in $R[[t, z]]$. Ordinary Lagrange–Bürmann inversion in z gives

$$U = \sum_{m \geq 1} \frac{z^m}{m} [u^{m-1}] \Psi(t, u)^m.$$

Because Ψ is divisible by t , only $m \leq n$ contributes to $[t^n]$; hence setting $z = 1$ is coefficientwise legitimate and gives Equation (6.5). Uniqueness also follows recursively in the t -adic topology. $\square$

6.2 Ordinary partial Bell polynomials

For a sequence $\gamma = (\gamma_1, \gamma_2, \dots)$, define

$$\widehat{B}_{n,k}(\gamma) := [t^n] \left(\sum_{r \geq 1} \gamma_r t^r \right)^k. \quad (6.6)$$

Thus $\widehat{B}_{0,0} = 1$, $\widehat{B}_{n,0} = 0$ for $n > 0$, and

$$\widehat{B}_{n,k}(\gamma) = \sum_{\substack{m_1, m_2, \dots \geq 0 \\ \sum m_r = k, \sum r m_r = n}} \frac{k!}{\prod_r m_r!} \prod_r \gamma_r^{m_r}. \quad (6.7)$$

These are ordinary, rather than exponential, partial Bell polynomials. They are the natural bookkeeping device for powers of a correction series whose coefficients are not divided by factorials.

7 Lambert-normalized inverse transseries

7.1 Exact inversion of the dominant block

Suppress Q in Equation (6.2). The equation

$$L = aX + b \log X \quad (7.1)$$

has, when $b \neq 0$, the exact Lambert form

$$X_k(L) = \frac{b}{a} W_k \left(\frac{a}{b} e^{L/b} \right). \quad (7.2)$$

For the swing parameters this becomes

$$X_{\sigma,k}(y) = \frac{\sigma}{2\lambda} W_k \left(2\sigma \lambda e^{2\sigma L_\sigma(y)} \right), \quad L_\sigma(y) = \log \frac{y}{C_\sigma}. \quad (7.3)$$

The real large branches are

$$\boxed{X_+(y) = \frac{W_0(4\pi\lambda y^2)}{2\lambda}, \quad X_-(y) = -\frac{W_{-1}(-4\lambda/(\pi y^2))}{2\lambda}.} \quad (7.4)$$

The appearance of W_{-1} in the even phase is essential. Its argument approaches zero through negative values, and $W_{-1}(z) \rightarrow -\infty$, which produces the large positive root. The principal branch W_0 would produce the small root of the simplified equation and is irrelevant to the large-index inverse.

7.2 All constant coefficients around the Lambert core

Theorem 7.1 (Lambert-centered reversion). *Let X solve Equation (7.1), let $t = X^{-1}$, and seek the inverse of Equation (6.2) in the form*

$$x = X + U(t), \quad U(t) = \sum_{n \geq 1} c_n t^n. \quad (7.5)$$

Define

$$\Psi(t, u) := -\frac{1}{a} \left[b \log(1 + tu) + Q\left(\frac{t}{1 + tu}\right) \right]. \quad (7.6)$$

Then $U = \Psi(t, U)$, and for every $n \geq 1$,

$$c_n = \sum_{m=1}^n \frac{1}{m} [t^n u^{m-1}] \Psi(t, u)^m. \quad (7.7)$$

Equivalently, put $\gamma_1 = 0$ and $\gamma_r = c_{r-1}$ for $r \geq 2$. Then the coefficients are generated triangularly by

$$\begin{aligned} c_n = -\frac{1}{a} \left[b \sum_{k=1}^n \frac{(-1)^{k+1}}{k} \widehat{B}_{n,k}(\gamma) \right. \\ \left. + \sum_{j=1}^n q_j \sum_{k=0}^{n-j} (-1)^k \binom{j+k-1}{k} \widehat{B}_{n-j,k}(\gamma) \right]. \end{aligned} \quad (7.8)$$

Only $c_1, \dots, c_{n-1}$ occur on the right-hand side.

Proof. Subtract $aX + b \log X = L$ from $a(X + U) + b \log(X + U) + Q((X + U)^{-1}) = L$. Since $X + U = X(1 + tU)$, the result is precisely $U = \Psi(t, U)$. Equation (7.7) is Theorem 6.1. To obtain Equation (7.8), write $w = tU = \sum_{r \geq 1} \gamma_r t^r$, expand

$$\log(1 + w) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} w^k, \quad (1 + w)^{-j} = \sum_{k \geq 0} (-1)^k \binom{j+k-1}{k} w^k,$$

and extract $[t^n]$. □

Remark 7.2. The dominant logarithm has been resummed exactly into X . Therefore the remaining c_n are constants in the coefficient ring; no powers of $\log L$ occur. This is the principal practical advantage over direct power–logarithmic reversion.

7.3 Specialization to the swing factorial

For Equation (6.3), define

$$\Psi_\sigma(t, u) = -\frac{1}{\lambda} \left[\frac{\sigma}{2} \log(1 + tu) + \sigma \sum_{j \geq 1} d_j \left(\frac{t}{1 + tu} \right)^j \right]. \quad (7.9)$$

Then

$$c_n^{(\sigma)} = \sum_{m=1}^n \frac{1}{m} [t^n u^{m-1}] \Psi_\sigma(t, u)^m. \quad (7.10)$$

The first coefficients, reduced with $\sigma^2 = 1$, are

$$c_1^{(\sigma)} = -\frac{\sigma}{4\lambda}, \quad c_2^{(\sigma)} = \frac{1}{8\lambda^2}, \quad (7.11)$$

$$c_3^{(\sigma)} = \frac{2\sigma\lambda^2 - 3\lambda - 3\sigma}{48\lambda^3}, \quad c_4^{(\sigma)} = \frac{-4\lambda^2 + 15\sigma\lambda + 6}{192\lambda^4}, \quad (7.12)$$

$$c_5^{(\sigma)} = -\frac{96\sigma\lambda^4 - 80\lambda^3 + 40\sigma\lambda^2 + 135\lambda + 30\sigma}{1920\lambda^5}, \quad (7.13)$$

$$c_6^{(\sigma)} = \frac{96\lambda^4 - 180\sigma\lambda^3 + 215\lambda^2 + 210\sigma\lambda + 30}{3840\lambda^6}. \quad (7.14)$$

Thus the odd inverse begins

$$\begin{aligned} \mathcal{I}_+(y) \sim X_+ - \frac{1}{4\lambda X_+} + \frac{1}{8\lambda^2 X_+^2} + \frac{2\lambda^2 - 3\lambda - 3}{48\lambda^3 X_+^3} \\ + \frac{-4\lambda^2 + 15\lambda + 6}{192\lambda^4 X_+^4} + \dots, \end{aligned} \quad (7.15)$$

and the even inverse begins

$$\begin{aligned} \mathcal{I}_-(y) \sim X_- + \frac{1}{4\lambda X_-} + \frac{1}{8\lambda^2 X_-^2} + \frac{-2\lambda^2 - 3\lambda + 3}{48\lambda^3 X_-^3} \\ + \frac{-4\lambda^2 - 15\lambda + 6}{192\lambda^4 X_-^4} + \dots. \end{aligned} \quad (7.16)$$

The sign of the first correction follows already from $D(x) > 0$: the odd phase has $+D$ in its logarithm and therefore reaches a fixed value at an index smaller than X_+ , while the even phase has $-D$ and reaches it at an index larger than X_- .

7.4 Complex inverse branches

Every Lambert branch in Equation (7.3) that satisfies $|X_{\sigma,k}| \rightarrow \infty$ in a sector supplies a formal complex inverse branch

$$\mathcal{I}_{\sigma,k}(y) \sim X_{\sigma,k}(y) + \sum_{n \geq 1} c_n^{(\sigma)} X_{\sigma,k}(y)^{-n}. \quad (7.17)$$

The coefficient sequence is independent of k ; all branch dependence is carried by W_k and by the chosen logarithm in L_σ . Analytic validity requires that the path remain inside a sector avoiding poles of the gamma quotient and Stokes directions of its Borel transform. The standard branch structure and large-argument expansion of W_k are described in [6, 7].

8 Direct power-logarithmic inverse transseries

Lambert normalization is compact and numerically efficient, but it hides the explicit polynomial-logarithmic scale. We now expand it completely. For the general model Equation (6.1), put

$$L = \log(y/C), \quad H = \log(L/a). \quad (8.1)$$

We order monomials lexicographically by powers of L^{-1} , with $H = \mathcal{O}(\log L)$ treated as a slow variable.

8.1 Universal coefficient formula

Theorem 8.1 (Direct reversion at infinity). *The formal inverse of Equation (6.2) has the unique form*

$$x \sim \frac{1}{a} \left[L - bH + \sum_{n \geq 1} \frac{P_n(H)}{L^n} \right]. \quad (8.2)$$

Let

$$\Phi(t, u; H) := -b \log(1 + t(u - bH)) - Q\left(\frac{at}{1 + t(u - bH)}\right). \quad (8.3)$$

Then

$$P_n(H) = \sum_{m=1}^n \frac{1}{m} [t^n u^{m-1}] \Phi(t, u; H)^m. \quad (8.4)$$

For a triangular Bell recurrence, set

$$\gamma_1 = -bH, \quad \gamma_r = P_{r-1}(H) \quad (r \geq 2). \quad (8.5)$$

Then

$$\begin{aligned} P_n(H) = & -b \sum_{k=1}^n \frac{(-1)^{k+1}}{k} \hat{B}_{n,k}(\gamma) \\ & - \sum_{j=1}^n q_j a^j \sum_{k=0}^{n-j} (-1)^k \binom{j+k-1}{k} \hat{B}_{n-j,k}(\gamma). \end{aligned} \quad (8.6)$$

In particular P_n has degree at most n . If $b \neq 0$, its leading term is

$$P_n(H) = \frac{b^{n+1}}{n} H^n + \text{terms of degree at most } n-1. \quad (8.7)$$

Proof. Set $t = L^{-1}$ and $V(t) = \sum_{n \geq 1} P_n(H) t^n$. The ansatz gives

$$ax = t^{-1} - bH + V, \quad x = \frac{1}{at} (1 + t(V - bH)).$$

Since $\log(1/(at)) = H$, substitution in $t^{-1} = ax + b \log x + Q(x^{-1})$ cancels the two occurrences of bH and leaves

$$V = \Phi(t, V; H).$$

Equation (8.4) follows from Theorem 6.1. Writing $w = t(V - bH) = \sum_{r \geq 1} \gamma_r t^r$, then expanding $\log(1 + w)$ and $(1 + w)^{-j}$, gives Equation (8.6). Induction proves the degree bound. The highest power of H is unaffected by Q ; extracting it from the pure logarithmic equation gives Equation (8.7), or follows immediately from Theorem 8.2 below. $\square$

8.2 Closed Stirling formula for the pure Lambert block

Let $\left[\begin{smallmatrix} n \\ m \end{smallmatrix} \right]$ denote the unsigned Stirling number of the first kind.

Theorem 8.2 (Pure Lambert coefficients). *When $Q = 0$, the polynomials in Equation (8.2) are*

$$P_n^{(0)}(H) = \frac{b^{n+1}}{n!} \sum_{m=1}^n (-1)^{m+1} (m-1)! \left[\begin{smallmatrix} n \\ m \end{smallmatrix} \right] \binom{n}{m-1} H^{n-m+1}. \quad (8.8)$$

Proof. With $Q = 0$, Equation (8.4) becomes

$$P_n^{(0)} = \sum_{m=1}^n \frac{(-b)^m}{m} [t^n u^{m-1}] \log(1 + t(u - bH))^m.$$

Use the signed Stirling expansion

$$\log(1 + z)^m = m! \sum_{n \geq m} s(n, m) \frac{z^n}{n!}, \quad s(n, m) = (-1)^{n-m} \begin{bmatrix} n \\ m \end{bmatrix}.$$

The coefficient of $t^n u^{m-1}$ is obtained from $(u - bH)^n$. Simplifying the signs gives Equation (8.8). $\square$

The first pure terms are

$$\begin{aligned} P_1^{(0)} &= b^2 H, \\ P_2^{(0)} &= \frac{b^3}{2} (H^2 - 2H), \\ P_3^{(0)} &= \frac{b^4}{6} (2H^3 - 9H^2 + 6H), \\ P_4^{(0)} &= \frac{b^5}{12} (3H^4 - 22H^3 + 36H^2 - 12H). \end{aligned}$$

Formula (8.8) is the precise bridge between Lambert asymptotics and the first-kind Stirling coefficient calculus.

8.3 Explicit swing inverse polynomials

For the swing factorial, let

$$L_\sigma = \log y + \frac{1}{2} \log \pi + \frac{\sigma}{2} \lambda, \quad H_\sigma = \log \frac{L_\sigma}{\lambda}. \quad (8.9)$$

Then

$$\boxed{\mathcal{I}_\sigma(y) \sim \frac{1}{\lambda} \left[L_\sigma - \frac{\sigma}{2} H_\sigma + \sum_{n \geq 1} \frac{P_n^{(\sigma)}(H_\sigma)}{L_\sigma^n} \right]}. \quad (8.10)$$

The first five coefficient polynomials are

$$P_1^{(\sigma)}(H) = \frac{H - \sigma\lambda}{4}, \quad (8.11)$$

$$P_2^{(\sigma)}(H) = \frac{\sigma H^2 - 2(\lambda + \sigma)H + 2\lambda}{16}, \quad (8.12)$$

$$\begin{aligned} P_3^{(\sigma)}(H) &= \frac{1}{96} \left[2H^3 - (6\sigma\lambda + 9)H^2 + (18\sigma\lambda + 6)H \right. \\ &\quad \left. + 4\sigma\lambda^3 - 6\lambda^2 - 6\sigma\lambda \right], \end{aligned} \quad (8.13)$$

$$\begin{aligned} P_4^{(\sigma)}(H) &= \frac{1}{384} \left[3\sigma H^4 - (12\lambda + 22\sigma)H^3 + (66\lambda + 36\sigma)H^2 \right. \\ &\quad \left. + (24\lambda^3 - 36\sigma\lambda^2 - 72\lambda - 12\sigma)H \right. \\ &\quad \left. - 8\lambda^3 + 30\sigma\lambda^2 + 12\lambda \right], \end{aligned} \quad (8.14)$$

$$P_5^{(\sigma)}(H) = \frac{1}{3840} \left[12H^5 - (60\sigma\lambda + 125)H^4 + (500\sigma\lambda + 350)H^3 \right.$$

$$\begin{aligned}
& + (240\sigma\lambda^3 - 360\lambda^2 - 1050\sigma\lambda - 300)H^2 \\
& + (-280\sigma\lambda^3 + 780\lambda^2 + 600\sigma\lambda + 60)H \\
& - 192\sigma\lambda^5 + 160\lambda^4 - 80\sigma\lambda^3 - 270\lambda^2 - 60\sigma\lambda \Big]. \tag{8.15}
\end{aligned}$$

Every coefficient is obtained by the single recurrence Equation (8.6) with $b = \sigma/2$ and $q_j = \sigma d_j$; no order-specific ansatz is needed.

8.4 Relation between the two inverse normalizations

The Lambert-centered expansion Equation (7.5) resums the entire pure block Equation (8.8). Expanding W_0 or W_{-1} in iterated logarithms and then expanding each power X^{-n} reproduces Equation (8.10). Conversely, grouping in the direct series all terms generated solely by $(\sigma/2) \log x$ reconstructs X_σ .

Formally, the direct coefficients live in

$$\mathbb{Q}[\lambda, \lambda^{-1}, \sigma, H]/(\sigma^2 - 1),$$

whereas the Lambert-centered coefficients live in the smaller constant ring

$$\mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1).$$

This algebraic compression explains the substantially better numerical behavior observed in Section 11.

9 Perturbative Lagrange operators and exact inverse transport

There is a third all-orders description that is especially convenient for Borel and Stokes analysis.

Theorem 9.1 (Inverse perturbation operator). *Let F_0 be locally invertible, let $X = F_0^{-1}(L)$, and consider*

$$L = F_0(x) + \varepsilon R(x). \tag{9.1}$$

Define the derivative with respect to the leading coordinate by

$$\mathcal{D}_0 := \frac{1}{F'_0(X)} \frac{d}{dX}. \tag{9.2}$$

Then, as a formal power series in ε ,

$$x = X + \sum_{m \geq 1} \frac{(-\varepsilon)^m}{m!} \mathcal{D}_0^{m-1} \left[\frac{R(X)^m}{F'_0(X)} \right]. \tag{9.3}$$

Proof. Use $u = F_0(x)$. Equation (9.1) becomes $L = u + \varepsilon r(u)$, where $r(u) = R(F_0^{-1}(u))$. Lagrange–Bürmann inversion for the composite function $F_0^{-1}(u)$ gives

$$F_0^{-1}(u(L)) = F_0^{-1}(L) + \sum_{m \geq 1} \frac{(-\varepsilon)^m}{m!} \left(\frac{d}{dL} \right)^{m-1} [(F_0^{-1})'(L) r(L)^m].$$

Now $(F_0^{-1})'(L) = 1/F'_0(X)$ and $d/dL = \mathcal{D}_0$. □

For the swing problem,

$$F_{0,\sigma}(x) = \lambda x + \frac{\sigma}{2} \log x, \quad R_\sigma(x) = \sigma D(x), \quad F'_{0,\sigma}(x) = \lambda + \frac{\sigma}{2x}. \quad (9.4)$$

Thus, with $X = X_\sigma(y)$,

$$\mathcal{I}_\sigma(y) = X + \sum_{m \geq 1} \frac{(-\sigma)^m}{m!} \mathcal{D}_{0,\sigma}^{m-1} \left[\frac{D(X)^m}{\lambda + \sigma/(2X)} \right], \quad (9.5)$$

where D may be interpreted either as its formal asymptotic series or as the exact Borel sum Equation (3.4). The first three perturbative layers are

$$\begin{aligned} \mathcal{I}_\sigma(y) = X - \sigma \frac{D(X)}{F'_{0,\sigma}(X)} + \frac{1}{2} \mathcal{D}_{0,\sigma} \left[\frac{D(X)^2}{F'_{0,\sigma}(X)} \right] \\ - \frac{\sigma}{6} \mathcal{D}_{0,\sigma}^2 \left[\frac{D(X)^3}{F'_{0,\sigma}(X)} \right] + \dots \end{aligned} \quad (9.6)$$

Expanding $D(X)$ in inverse powers of X reproduces exactly the coefficients $c_n^{(\sigma)}$.

10 Borel singularities, Stokes jumps, and inverse transseries

For a direction θ , write

$$\mathcal{S}_{\theta^\pm} D(x) := \int_0^{e^{i(\theta \pm 0)} \infty} e^{-xt} \widehat{D}(t) dt. \quad (10.1)$$

whenever the rotated Laplace integral converges. Define the discontinuity by lower minus upper lateral summation. The residue theorem and Equation (3.8) give the exact bridge equations

$$\text{Disc}_{\pi/2} D(x) := \mathcal{S}_{(\pi/2)^-} D - \mathcal{S}_{(\pi/2)^+} D = 2 \sum_{k \geq 0} \frac{e^{-i\pi(2k+1)x}}{2k+1}, \quad (10.2)$$

$$\text{Disc}_{-\pi/2} D(x) := \mathcal{S}_{(-\pi/2)^-} D - \mathcal{S}_{(-\pi/2)^+} D = -2 \sum_{k \geq 0} \frac{e^{i\pi(2k+1)x}}{2k+1}. \quad (10.3)$$

In their respective convergence half-planes these may be summed as

$$\text{Disc}_{\pi/2} D = 2 \operatorname{artanh}(e^{-i\pi x}), \quad \text{Disc}_{-\pi/2} D = -2 \operatorname{artanh}(e^{i\pi x}). \quad (10.4)$$

The logarithm of the swing branch has jump

$$\text{Disc}_\theta \log \mathcal{S}_\sigma(x) = \sigma \text{Disc}_\theta D(x), \quad (10.5)$$

so the function itself changes multiplicatively by the exponential of this quantity.

Proposition 10.1 (Transport of a Stokes jump through inversion). *Let $F(x)$ be one lateral logarithmic branch and let the other be $F(x) + \Delta(x)$. If $x = F^{-1}(L)$, the inverse on the second side has the formal expansion*

$$x_{\text{new}} = x + \sum_{m \geq 1} \frac{(-1)^m}{m!} \left(\frac{1}{F'(x)} \frac{d}{dx} \right)^{m-1} \left[\frac{\Delta(x)^m}{F'(x)} \right]. \quad (10.6)$$

In particular,

$$x_{\text{new}} - x = -\frac{\Delta(x)}{F'(x)} + \text{ex higher exponential sectors}. \quad (10.7)$$

For the swing inverse, take $\Delta = \sigma \text{Disc}_\theta D$ and $F' = \lambda + \sigma/(2x) + \sigma D'(x)$.

Proof. Apply Theorem 9.1 with $F_0 = F$, $\varepsilon = 1$, and $R = \Delta$. □

Equations (10.2), (10.3) and (10.6) complete the resurgent transseries data: the Borel singularity locations, residues, all lateral jumps, and their nonlinear transport through the inverse are explicit. Along the positive real axis the preferred sum is Equation (3.4); no lateral choice is necessary.

11 Error control, computation, and numerical validation

11.1 Residual certification

Asymptotic inversion should be checked in the original equation, preferably in logarithmic form to avoid overflow.

Proposition 11.1 (Residual-to-index error bound). *Let x_* approximate $\mathcal{I}_\sigma(y)$, and define*

$$r_* := \log \mathcal{S}_\sigma(x_*) - \log y. \quad (11.1)$$

If $(\log \mathcal{S}_\sigma)'(x) \geq m > 0$ on the interval joining x_ and $\mathcal{I}_\sigma(y)$, then*

$$|x_* - \mathcal{I}_\sigma(y)| \leq \frac{|r_*|}{m}. \quad (11.2)$$

Proof. Apply the mean-value theorem to $\log \mathcal{S}_\sigma$. □

The exact Laplace representation gives

$$0 < -D'(x) < \frac{1}{4x^2}, \quad (11.3)$$

so for $x \geq 1$,

$$(\log \mathcal{S}_+)'(x) > \lambda + \frac{1}{2x} - \frac{1}{4x^2} > \lambda, \quad (11.4)$$

$$(\log \mathcal{S}_-)'(x) > \lambda - \frac{1}{2x}. \quad (11.5)$$

These elementary bounds are sufficient for certified large-index inversion; tighter bounds are available directly from the digamma expression Equation (2.9).

11.2 Newton refinement

Starting from either asymptotic approximation, Newton's method applied to the logarithmic equation is

$$x_{j+1} = x_j - \frac{\log \mathcal{S}_\sigma(x_j) - \log y}{\lambda + \frac{\sigma}{2} [\psi(x_j/2 + 1) - \psi(x_j/2 + 1/2)]}. \quad (11.6)$$

Evaluation through $\log \Gamma$ and ψ is stable even when $\mathcal{S}_\sigma(x)$ itself is far outside floating-point range. Once the asymptotic seed is in its natural regime, one or two Newton steps usually recover full working precision.

11.3 Forward accuracy

Table 2 uses $x = 20$. The truncation labeled N retains $A_0, \dots, A_{N-1}$ in Equation (5.5).

The decrease is far faster than a fixed-order theorem alone suggests because $x = 20$ is still well before the least-term index, which is near $\pi x/2 \approx 31$ in the nonzero logarithmic series.

Table 2: Relative errors of the forward expansion at $x = 20$.

N	even branch $\sigma = -1$	odd branch $\sigma = +1$
1	1.2573193×10^{-2}	1.2417071×10^{-2}
2	8.3971506×10^{-5}	7.2284564×10^{-5}
4	7.9979454×10^{-8}	4.8170528×10^{-8}
6	$3.2525110 \times 10^{-10}$	$8.8967744 \times 10^{-11}$
8	$3.2108180 \times 10^{-12}$	$9.7020566 \times 10^{-14}$
10	$5.8884939 \times 10^{-14}$	$1.5606307 \times 10^{-14}$
12	$1.7866041 \times 10^{-15}$	$6.7178815 \times 10^{-16}$

11.4 Inverse accuracy and the advantage of Lambert centering

For each phase, set $y = \mathcal{S}_\sigma(50)$, so that the exact inverse is 50. In Table 3, $N = 0$ denotes only the leading core: $(L_\sigma - \sigma H_\sigma/2)/\lambda$ in the direct column and X_σ in the Lambert column. For $N > 0$, the first N correction terms are retained.

Table 3: Absolute inverse errors for an exact index $x = 50$.

N	even phase $\sigma = -1$		odd phase $\sigma = +1$	
	direct	Lambert	direct	Lambert
0	4.92291×10^{-2}	7.31859×10^{-3}	3.24899×10^{-2}	7.11042×10^{-3}
1	9.28054×10^{-4}	1.04056×10^{-4}	2.43401×10^{-4}	1.02029×10^{-4}
2	8.18595×10^{-6}	4.29703×10^{-8}	4.74681×10^{-6}	2.00947×10^{-6}
3	6.38122×10^{-7}	2.27690×10^{-8}	2.67151×10^{-7}	5.08233×10^{-8}
4	4.81875×10^{-8}	5.65588×10^{-11}	4.49650×10^{-9}	1.40431×10^{-9}
6	8.92594×10^{-12}	2.85289×10^{-13}	9.28838×10^{-12}	8.11044×10^{-13}
8	2.71054×10^{-13}	8.38506×10^{-16}	5.17449×10^{-15}	1.18162×10^{-15}
10	1.56412×10^{-16}	3.13520×10^{-18}	1.42623×10^{-17}	8.47275×10^{-19}

The nonmonotone error at a few low truncation orders is normal for an asymptotic series. The overall comparison confirms the structural point: Lambert centering has already resummed the largest tower of $(\log L)^m/L^n$ terms.

11.5 Recovering integer indices

For an approximate equation $\text{sf}(n) \approx y$, compute both continuous inverse branches and project to their parity lattices:

$$n_-^{\text{near}} = 2 \left\lfloor \frac{\mathcal{I}_-(y)}{2} + \frac{1}{2} \right\rfloor, \quad n_+^{\text{near}} = 2 \left\lfloor \frac{\mathcal{I}_+(y) - 1}{2} + \frac{1}{2} \right\rfloor + 1. \quad (11.7)$$

Evaluate $\log \text{sf}(n)$ at these candidates and their nearest same-parity neighbors, then choose according to the desired criterion. For a threshold query, the greatest same-parity indices satisfying $\text{sf}(n) \leq y$ are obtained from

$$n_-^{\leq} = 2 \left\lfloor \frac{\mathcal{I}_-(y)}{2} \right\rfloor, \quad n_+^{\leq} = 2 \left\lfloor \frac{\mathcal{I}_+(y) - 1}{2} \right\rfloor + 1, \quad (11.8)$$

with a one-step exact correction if roundoff places an approximation across a boundary.

12 Further consequences of the two-phase inverse

12.1 Separation of the two inverse branches

For a common value y , define

$$Y := \log(y\sqrt{\pi}), \quad h := \log(Y/\lambda). \quad (12.1)$$

Then $L_\sigma = Y + \sigma\lambda/2$. Subtracting the first terms of Equation (8.10) gives

$$\mathcal{I}_-(y) - \mathcal{I}_+(y) = \frac{h}{\lambda} - 1 + \frac{1}{2Y} + \mathcal{O}\left(\frac{h^2}{Y^2}\right). \quad (12.2)$$

In particular,

$$\mathcal{I}_-(y) - \mathcal{I}_+(y) = \log_2\left(\frac{\log(y\sqrt{\pi})}{\log 2}\right) - 1 + o(1). \quad (12.3)$$

Thus the even index producing a given large magnitude exceeds the odd index by a slowly growing $\log \log y$ distance. This is the inverse reflection of the factor of order $x/2$ between the two forward branches at a common argument.

12.2 The counting function

Let

$$N(y) := \#\{n \in \mathbb{N}_0 : \text{sf}(n) \leq y\}. \quad (12.4)$$

Each parity subsequence is monotone, so

$$N(y) = \frac{\mathcal{I}_-(y) + \mathcal{I}_+(y)}{2} + \mathcal{O}(1). \quad (12.5)$$

The $\pm(1/2) \log x$ contributions cancel to leading order in the average, and therefore

$$N(y) = \frac{\log(y\sqrt{\pi})}{\log 2} + \mathcal{O}(1) = \log_2(y\sqrt{\pi}) + \mathcal{O}(1). \quad (12.6)$$

A smooth version of the counting function can be expanded to arbitrary order by averaging the two polynomial families $P_n^{(-)}$ and $P_n^{(+)}$ after substituting $L_\sigma = Y + \sigma\lambda/2$.

13 Symbolic coefficient generation

The following SymPy program implements the formulae of the article. It constructs d_j , the forward Bell coefficients A_n , the Lambert-centered coefficients $c_n^{(\sigma)}$, and the direct polynomials $P_n^{(\sigma)}(H)$. Reduction modulo $\sigma^2 - 1$ keeps the two parity phases in a single expression.

Listing 1: All-order coefficient generation.

```
import sympy as sp

N = 12
t, u, H = sp.symbols("t u H")
lam, sigma = sp.symbols("lambda sigma", nonzero=True)

# Logarithmic coefficients d_j; absent even coefficients are zero.
```

```

d = {
    2*r + 1:
        (2**(2*r + 2) - 1) * sp.bernoulli(2*r + 2)
        / ((2*r + 1) * (2*r + 2))
    for r in range(N)
}

D = sum(d.get(j, 0) * t**j for j in range(1, N + 1))
E = sp.series(sp.exp(D), t, 0, N + 1).removeO().expand()
A = [sp.factor(E.coeff(t, n)) for n in range(N + 1)]

def parity_reduce(expr):
    """Reduce a polynomial expression modulo sigma**2 - 1."""
    poly = sp.Poly(sp.expand(expr), sigma)
    ans = 0
    for (power,), coefficient in poly.terms():
        ans += coefficient * (sigma if power % 2 else 1)
    return sp.factor(ans)

# Lambert-centered inverse: U = Psi(t,U), U = sum c_n t^n.
U = sp.Integer(0)
c = []
for n in range(1, N + 1):
    Q_at_shift = sigma * sum(
        d.get(j, 0) * (t / (1 + t*U))**j
        for j in range(1, N + 1)
    )
    Psi = -(sigma*sp.log(1 + t*U)/2 + Q_at_shift) / lam
    cn = sp.series(Psi, t, 0, n + 1).removeO().expand().coeff(t, n)
    cn = parity_reduce(cn)
    c.append(cn)
    U += cn * t**n

# Direct inverse: V = Phi(t,V;H), V = sum P_n(H)t^n.
V = sp.Integer(0)
P = []
for n in range(1, N + 1):
    w = t * (V - sigma*H/2)
    Q_at_direct_shift = sigma * sum(
        d.get(j, 0) * (lam*t / (1 + w))**j
        for j in range(1, N + 1)
    )
    Phi = -sigma*sp.log(1 + w)/2 - Q_at_direct_shift
    pn = sp.series(Phi, t, 0, n + 1).removeO().expand().coeff(t, n)
    pn = parity_reduce(pn)
    P.append(pn)
    V += pn * t**n

for n, value in enumerate(A):
    print("A", n, "=", value)
for n, value in enumerate(c, 1):
    print("c", n, "=", value)
for n, value in enumerate(P, 1):
    print("P", n, "=", value)

```

The recurrence form is usually faster than direct multivariate coefficient extraction. At order N , truncated-series arithmetic gives a straightforward $\tilde{O}(N^2)$ implementation; Newton iteration for formal power series can reduce this further when thousands of terms are required.

14 Conclusions

The swing factorial is a compact example in which several asymptotic structures that are often studied separately occur at once.

1. Its parity split is leading-order information. The correct forward object is a two-component periodic transseries, and the correct inverse is a pair of monotone branches.
2. Duplication reduces both branches to one gamma ratio. After removal of its square-root scale, the entire logarithmic correction has the exact Borel transform $\tanh(t/2)/(2t)$.
3. Bernoulli numbers give the logarithmic coefficients; complete Bell polynomials exponentiate them; ordinary Bell polynomials organize nonlinear reversion; and unsigned first-kind Stirling numbers give the pure Lambert inverse block in closed form.
4. The odd real inverse is normalized by W_0 , whereas the even real inverse is normalized by W_{-1} . Missing this branch distinction selects the wrong root before any asymptotic correction is computed.
5. The Borel pole lattice supplies exact remainder signs, optimal truncation, lateral Stokes jumps, and, through the inverse perturbation operator, the full exponential-sector transport of every inverse branch.

All coefficients in both forward and inverse expansions are therefore computable by explicit finite formulae at arbitrary order. No unspecified “successive matching” step remains: each layer is either a Bernoulli number, a Bell polynomial, a Stirling number, or a coefficient extraction from a stated formal fixed-point equation.

A Additional Lambert-centered coefficients

For completeness, the next four coefficients after Equations (7.13) and (7.14) are

$$c_7^{(\sigma)} = \frac{1}{53760\lambda^7} \left(8160\sigma\lambda^6 - 4312\lambda^5 + 1428\sigma\lambda^4 + 1050\lambda^3 - 4025\sigma\lambda^2 - 2100\lambda - 210\sigma \right), \quad (\text{A.1})$$

$$c_8^{(\sigma)} = \frac{1}{645120\lambda^8} \left(-48960\lambda^6 + 56056\sigma\lambda^5 - 40908\lambda^4 + 15015\sigma\lambda^3 + 50610\lambda^2 + 17010\sigma\lambda + 1260 \right), \quad (\text{A.2})$$

$$c_9^{(\sigma)} = -\frac{1}{1290240\lambda^9} \left(1111040\sigma\lambda^8 - 413184\lambda^7 + 79616\sigma\lambda^6 + 43512\lambda^5 - 83748\sigma\lambda^4 + 83475\lambda^3 + 92610\sigma\lambda^2 + 22050\lambda + 1260\sigma \right), \quad (\text{A.3})$$

$$c_{10}^{(\sigma)} = \frac{1}{5160960\lambda^{10}} \left(2222080\lambda^8 - 1756032\sigma\lambda^7 + 779152\lambda^6 - 203280\sigma\lambda^5 - 162582\lambda^4 + 487305\sigma\lambda^3 + 309960\lambda^2 + 55440\sigma\lambda + 2520 \right). \quad (\text{A.4})$$

B A compact derivation from the general gamma-ratio expansion

The classical Tricomi–Erdélyi expansion for a ratio of gamma functions [4, 5] implies

$$\log \frac{\Gamma(z+a)}{\Gamma(z+b)} \sim (a-b) \log z + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n(n+1)z^n} (B_{n+1}(a) - B_{n+1}(b)), \quad (\text{B.1})$$

where $B_n(x)$ are Bernoulli polynomials. Taking $a = 1$, $b = 1/2$, and $z = x/2$, then using

$$B_m(1/2) = (2^{1-m} - 1)B_m, \quad B_{2r+1} = 0 \quad (r \geq 1),$$

reduces Equation (B.1) to Equation (4.3). The Laplace proof in the main text is stronger for this special ratio because it simultaneously supplies the exact Borel transform and remainder sign.

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